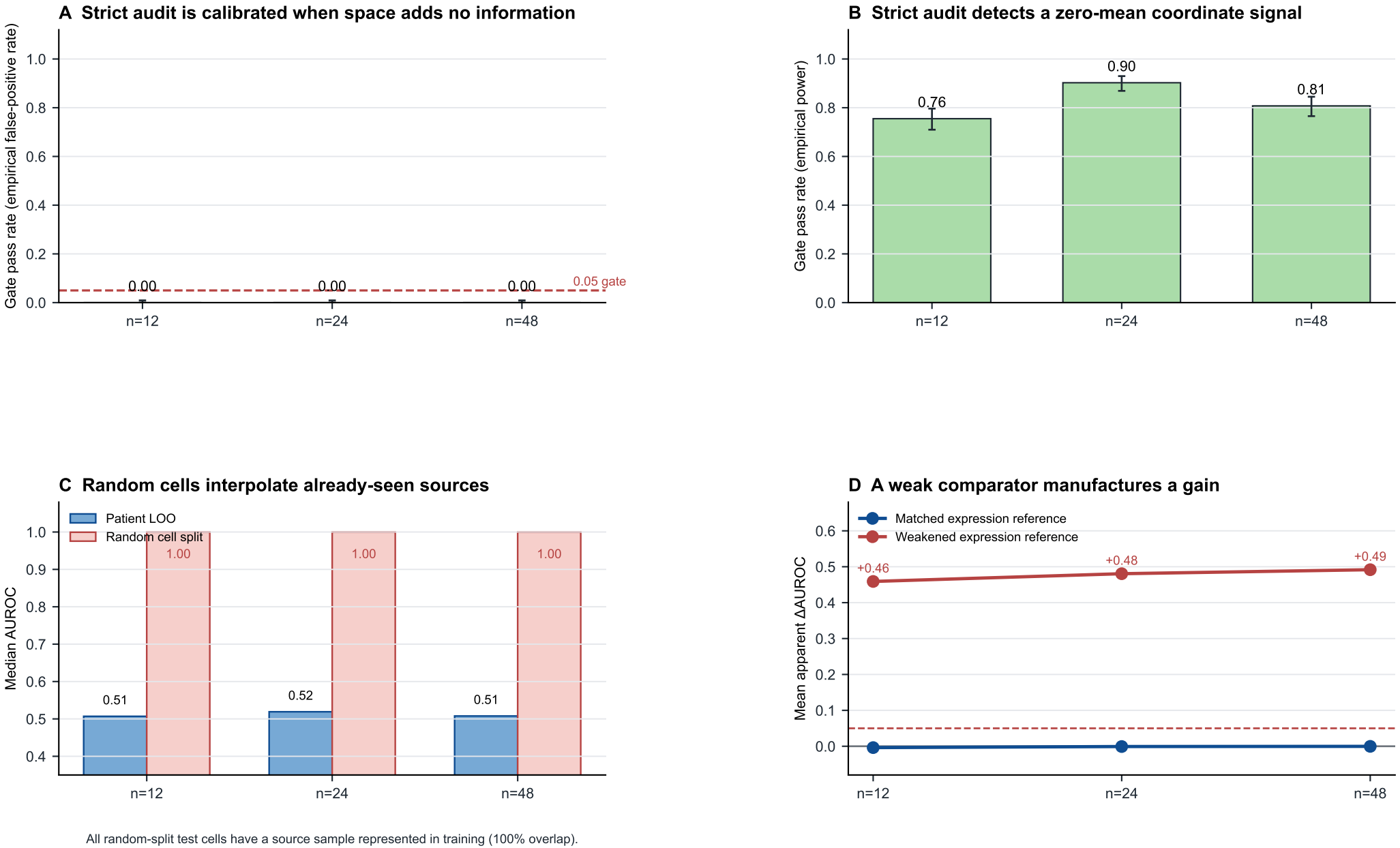

All random-split test cells have a source sample represented in training (100% overlap).

Each replicate contains paired patient samples and nested cells/spots. The strict gate requires ΔAUROC ≥ 0.05 and a patient-bootstrap lower 95% bound > 0. Error bars in A–B are exact 95% binomial intervals over Monte-Carlo replicates.